HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Python** scripts for automating data ingestion and generating scheduled reports, processing approximately **15 GB** of data daily.
- Designed and implemented **Java Spring Boot** REST APIs for inter-service communication, handling up to **500 requests per second** during peak loads.
- Contributed to the development of an enterprise **Java** application by implementing new features and resolving defects using **JUnit** and **Mockito**.
- Assisted in the deployment and management of containerized applications using **Docker** on **AWS EC2** instances, managing a fleet of **~30** servers.
- Wrote **SQL** queries and performed data validation for schema changes across **PostgreSQL** and **MySQL** databases, supporting **3** distinct enterprise systems.
- Managed production incidents for enterprise applications, performing root cause analysis using **Splunk** and ensuring adherence to SLAs for **P1/P2** issues.
- Collaborated with cross-functional teams to gather technical requirements and estimate effort for new feature development within an **Agile/Scrum** framework.

SKILLS

Languages – Python, Java, SQL, C/C++, Verilog, VHDL, Tcl, Perl, Shell Scripting

Frameworks & Libraries – Spring Boot, Hibernate, JPA, scikit-learn, pandas, NumPy, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile/Scrum, Linux, Splunk, Dynatrace, FPGA

Domain Knowledge – Embedded Software Design, Digital Design, Analog Sensing, High-Speed Connectivity, Security Algorithms, Wireless/Wired Communications

PROJECTS

Embedded Linux System Automation with Python

- Developed a suite of **Python** scripts to automate the deployment and configuration of embedded Linux systems across a fleet of **~50** development boards.
- Integrated **shell scripting** and **cron jobs** to manage system health checks and log aggregation, reducing manual intervention by **~40%**.
- Created interactive command-line tools using **FastAPI** for simplified system management and status reporting.

High-Speed Connectivity IP Validation Framework

- Designed and implemented validation test benches using **Verilog** and **C++** for a high-speed connectivity IP core, covering critical data path logic.
- Developed **Python** scripts to parse simulation logs and generate automated test reports, identifying over **15** critical bugs during the validation cycle.
- Contributed to the development of **FPGA** prototypes to accelerate verification cycles and test real-world scenarios.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)